SIMATS ENGINEERING

CO3 - ASSESSMENT TOOL - 1: Experiments

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA1522

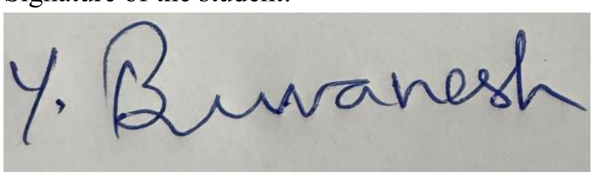
Name	Buvanesh Y
Register Number	192372367

COURSE OUTCOME COVEREDCO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)
Assessment Tool Weightage: 40%

Dave's Taxonomy: Precision (Level 3) **Total Marks: 40**
Question Count: 4

Duration : 60 Minutes

Code of Conduct
I **Buvanesh Y (192372367)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student:



Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

EXP NO 1: CREATE A SIMPLE CLOUD SOFTWARE APPLICATION FOR LIBRARY BOOK RESERVATION SYSTEM USING ANY CLOUD SERVICE PROVIDER TO DEMONSTRATE SaaS

AIM:

To create a simple cloud software application for Library Book Reservation System using Zoho Creator as a cloud service provider to demonstrate Software as a Service (SaaS).

PROCEDURE:

Step 1: Go to zoho.com.

Step 2: Log in to the Zoho account.

Step 3: Select Zoho Creator from the available applications.

Step 4: Create a new application and enter the application name as SIMATS Library Book Reservation System.

Step 5: Create the new application for managing library books.

Step 6: Select Form to create a book details form.

Step 7: Add the necessary fields to the form such as:

- Title of the Book
- Author
- Publication
- Year
- Edition
- No. of Copies
- Rack No.
- Status (Taken/Returned)

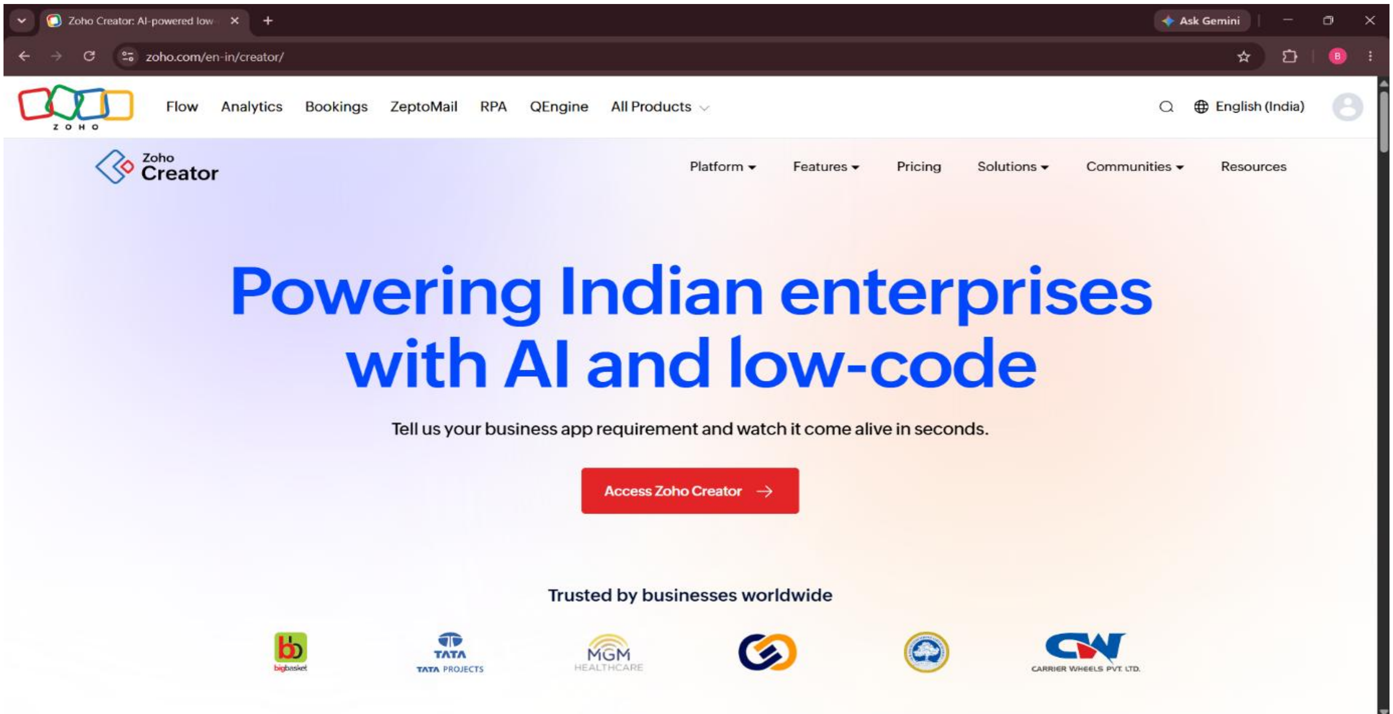
Step 8: Save the form and enter sample book details.

Step 9: Create a report to display all the available book details and their current status.

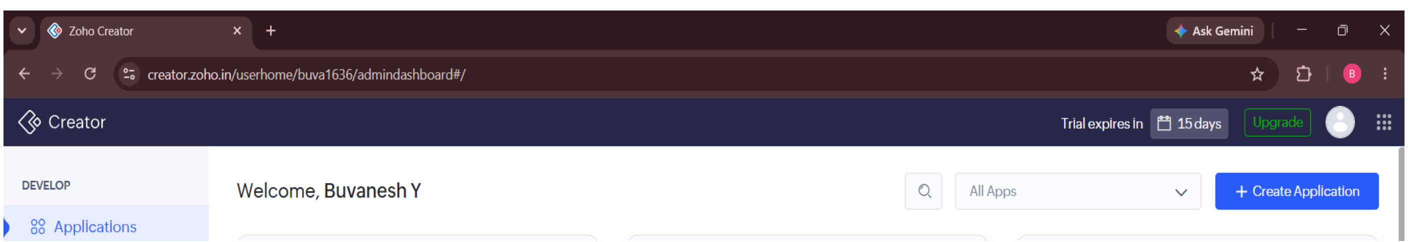
Step 10: The cloud-based library reservation software is created successfully and can be accessed through the internet.

IMPLEMENTATION:

STEP1: GOTO ZOHO.COM

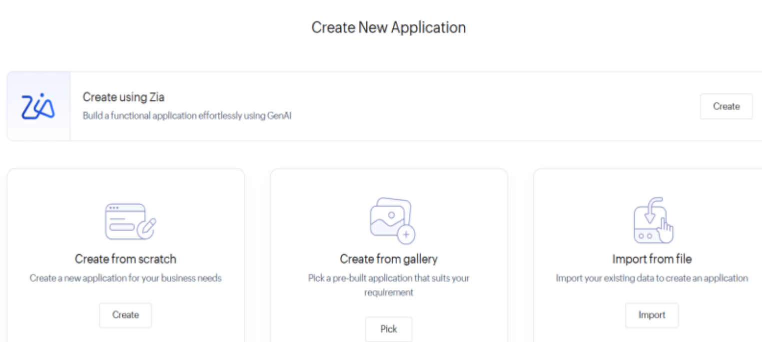


STEP 2: LOGINTOTHE ZOHO.COM

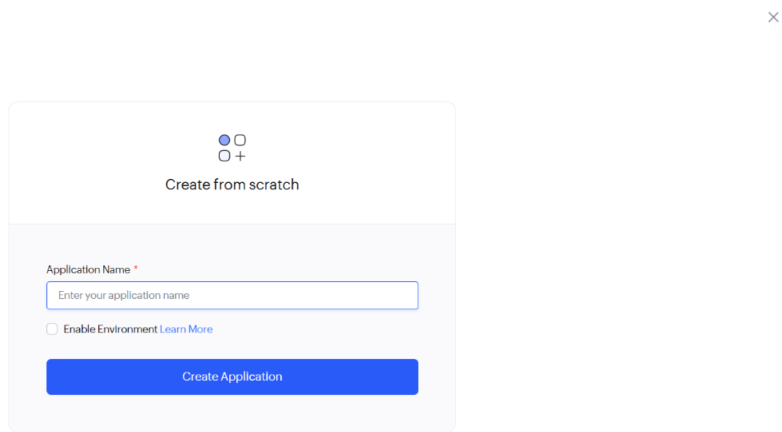


STEP 3: SELECT ONE APPLICATION

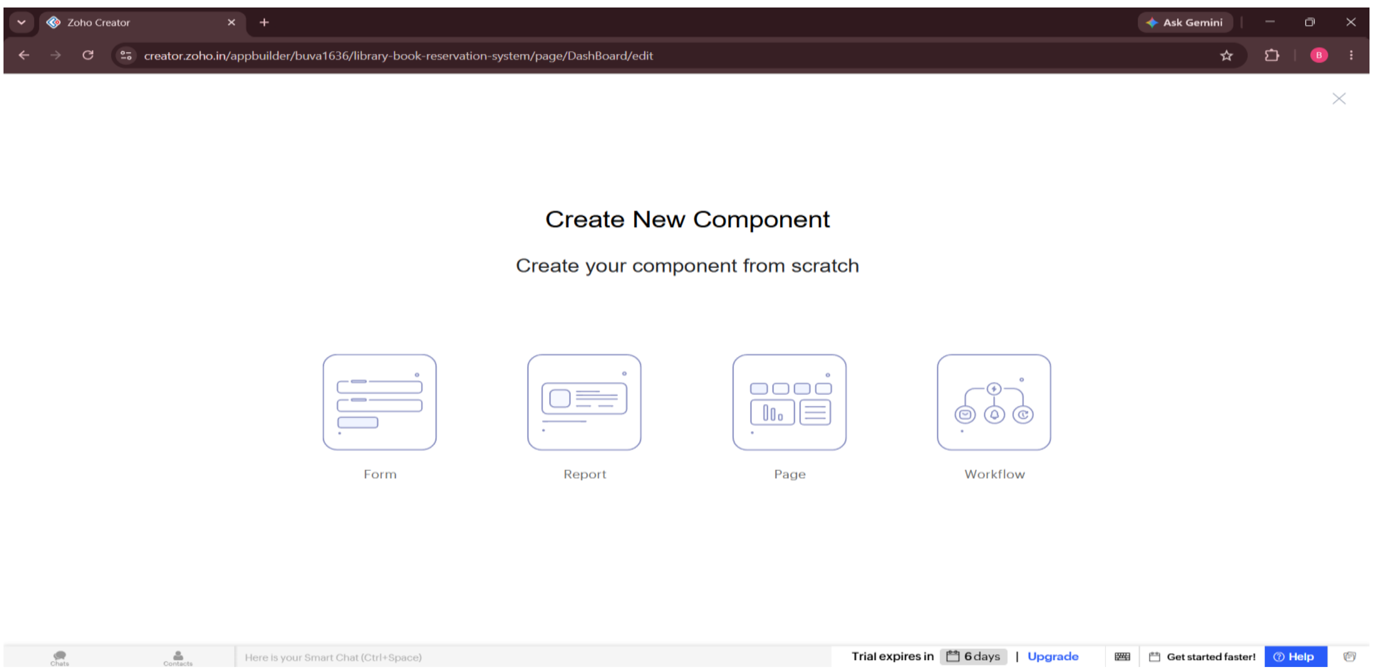
❖ Click Create from Scratch



STEP 4: ENTER APPLICATION NAME

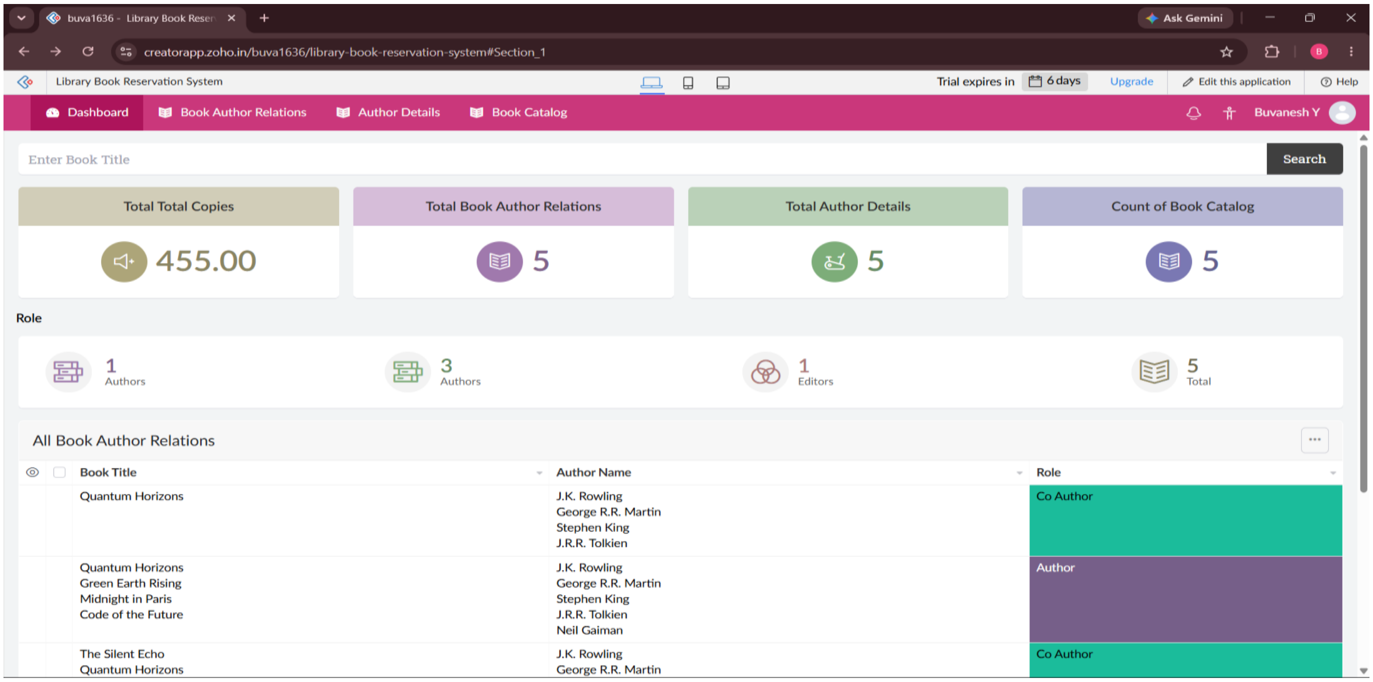


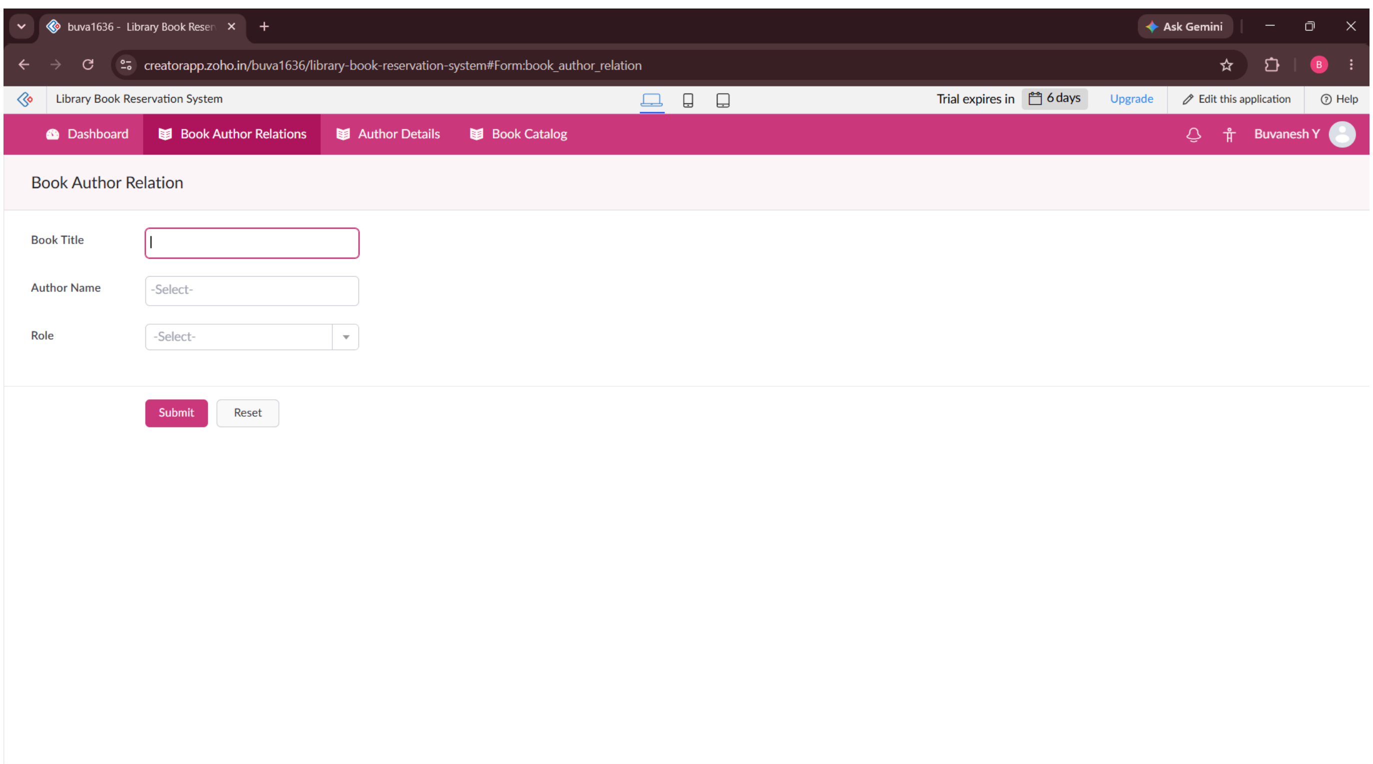
STEP 5: CREATED NEW APPLICATION

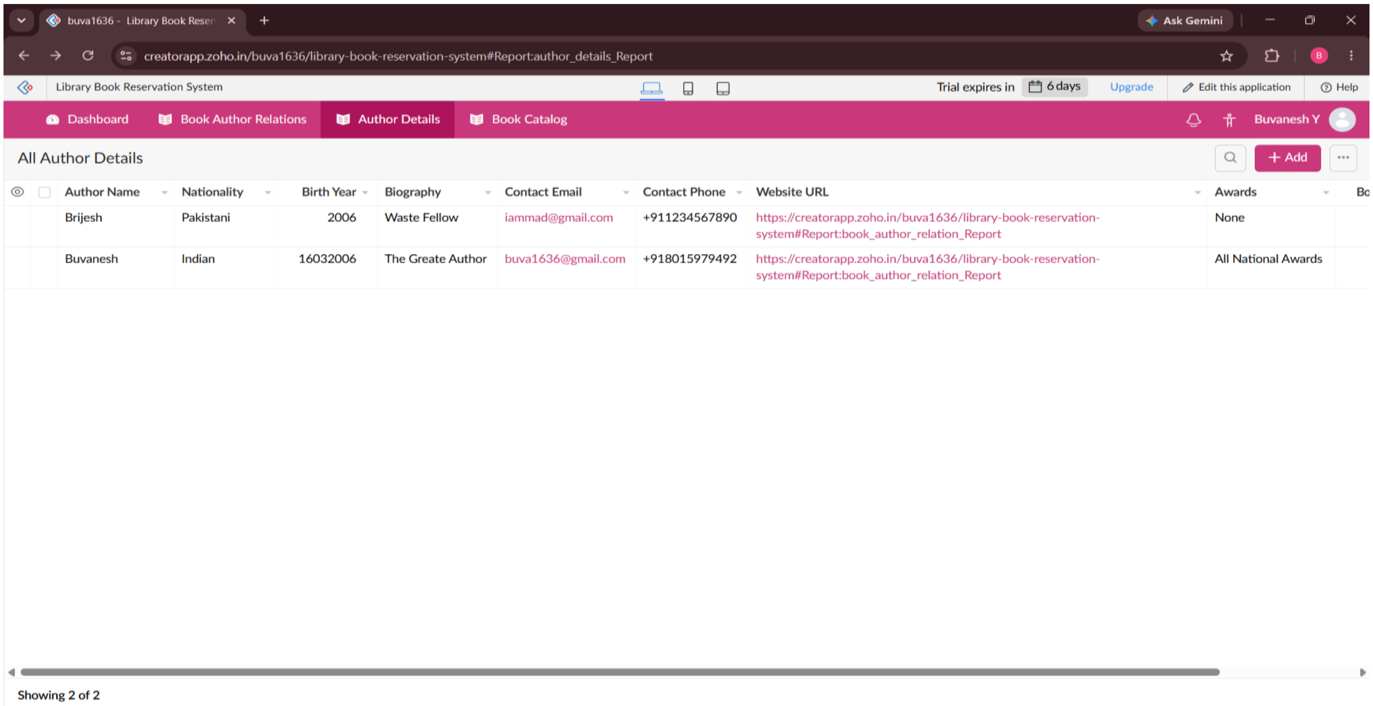


❖ Create Multiple Forms according to our application.

STEP 6: THE SOFTWARE HASE BEEN CREATED.







RESULT:

Thus, the **SIMATS Library Book Reservation System** was successfully created using **Zoho Creator** as a cloud service provider, demonstrating the **SaaS model**. The application allows library book details to be stored, managed, and accessed through the cloud.

EXP NO 2: CREATE A SIMPLE CLOUD SOFTWARE APPLICATION FOR PAYROLL PROCESSING SYSTEM USING ANY CLOUD SERVICE PROVIDER TO DEMONSTRATE SaaS

AIM:

To create a simple cloud software application for **Payroll Processing System** using **Zoho Creator** as a cloud service provider to demonstrate **Software as a Service (SaaS)**.

PROCEDURE:

Step 1: Go to **zoho.com**.

Step 2: Log in to the Zoho account.

Step 3: Select **Zoho Creator** from the available applications.

Step 4: Create a new application and enter the application name as **Payroll Processing System**.

Step 5: Create the new application for managing employee payroll details.

Step 6: Select **Form** to create an employee payroll details form.

Step 7: Add the necessary fields to the form such as:

- Name of the Employee
- Employee ID
- Experience in Present Organization
- Overall Experience
- Basic Pay
- HRA
- DA
- TA
- Gross Pay
- Net Pay

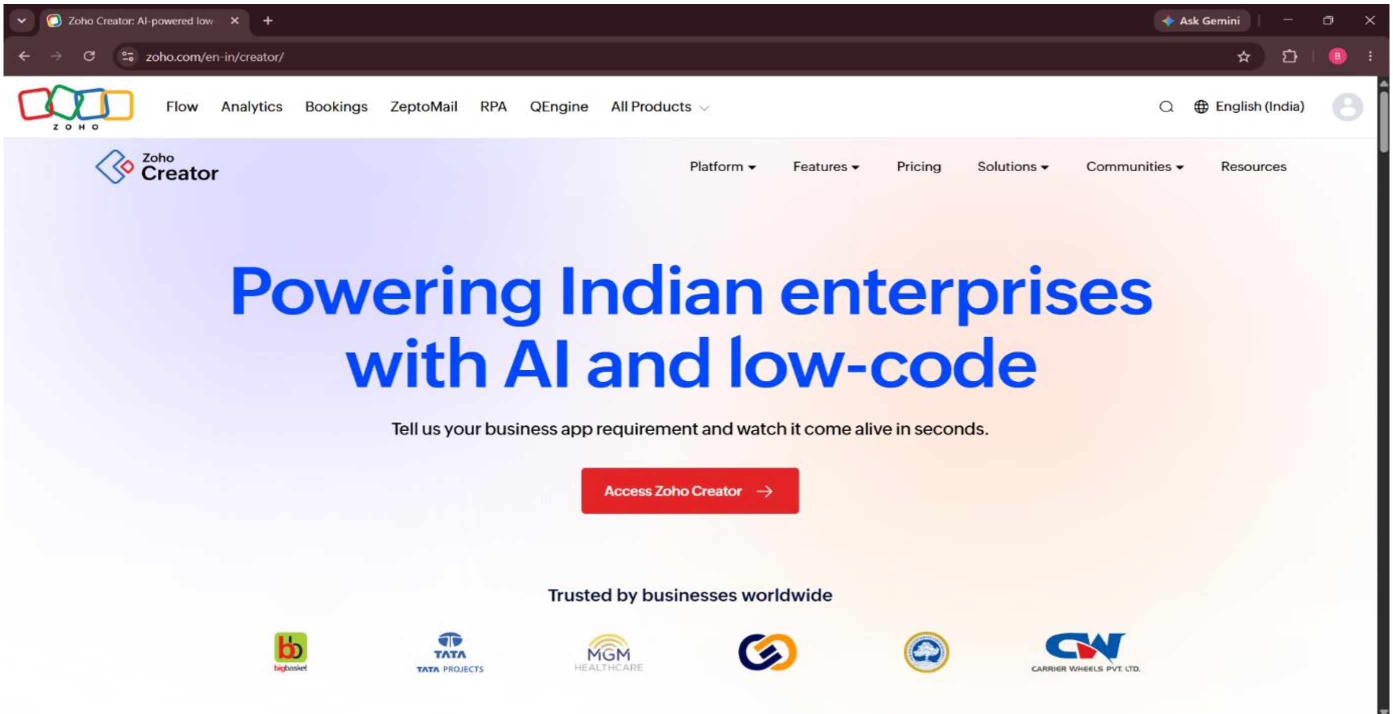
Step 8: Save the form and enter sample employee payroll details.

Step 9: Create a report to display the employee details and salary information.

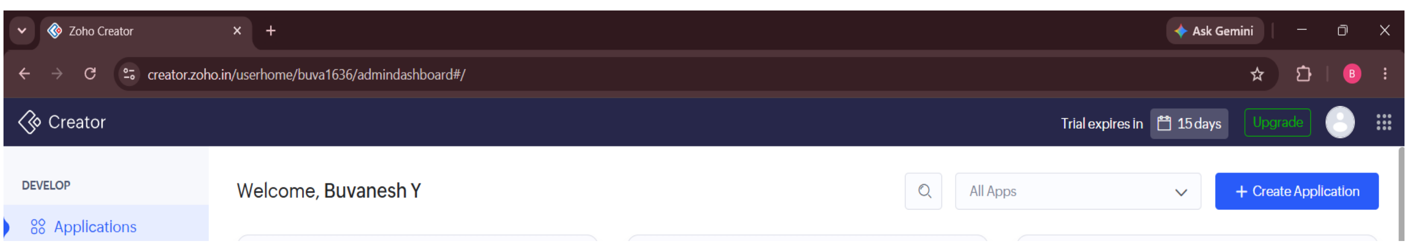
Step 10: The cloud-based Payroll Processing System is created successfully and can be accessed through the internet.

IMPLEMENTATION:

STEP1: GOTO ZOHO.COM

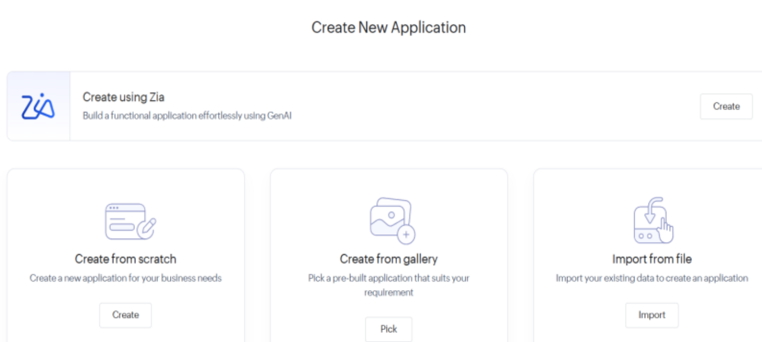


STEP 2: LOGINTOTHE ZOHO.COM

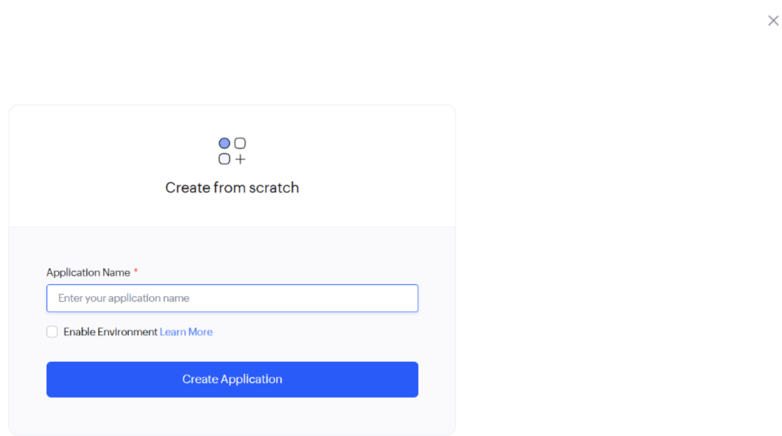


STEP 3: SELECT ONE APPLICATION

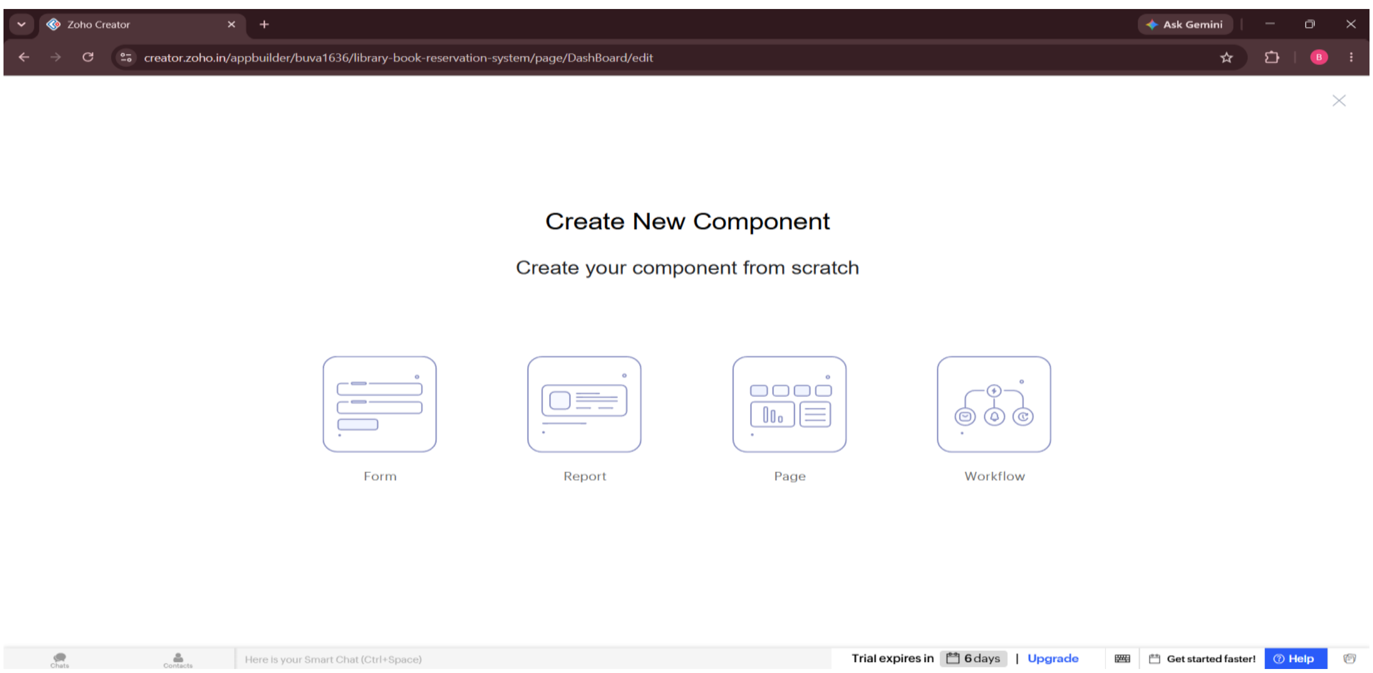
- ❖ Click Create from Scratch



STEP 4: ENTER APPLICATION NAME

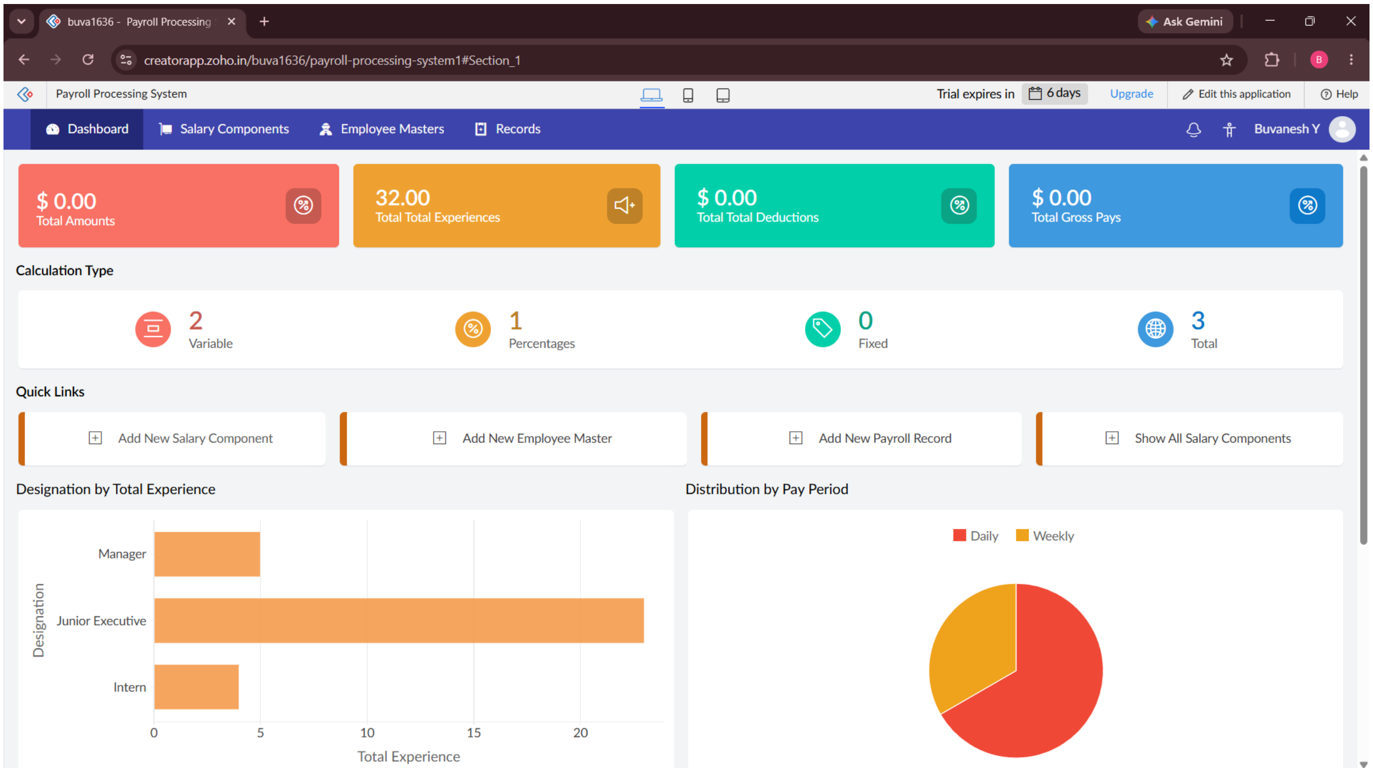


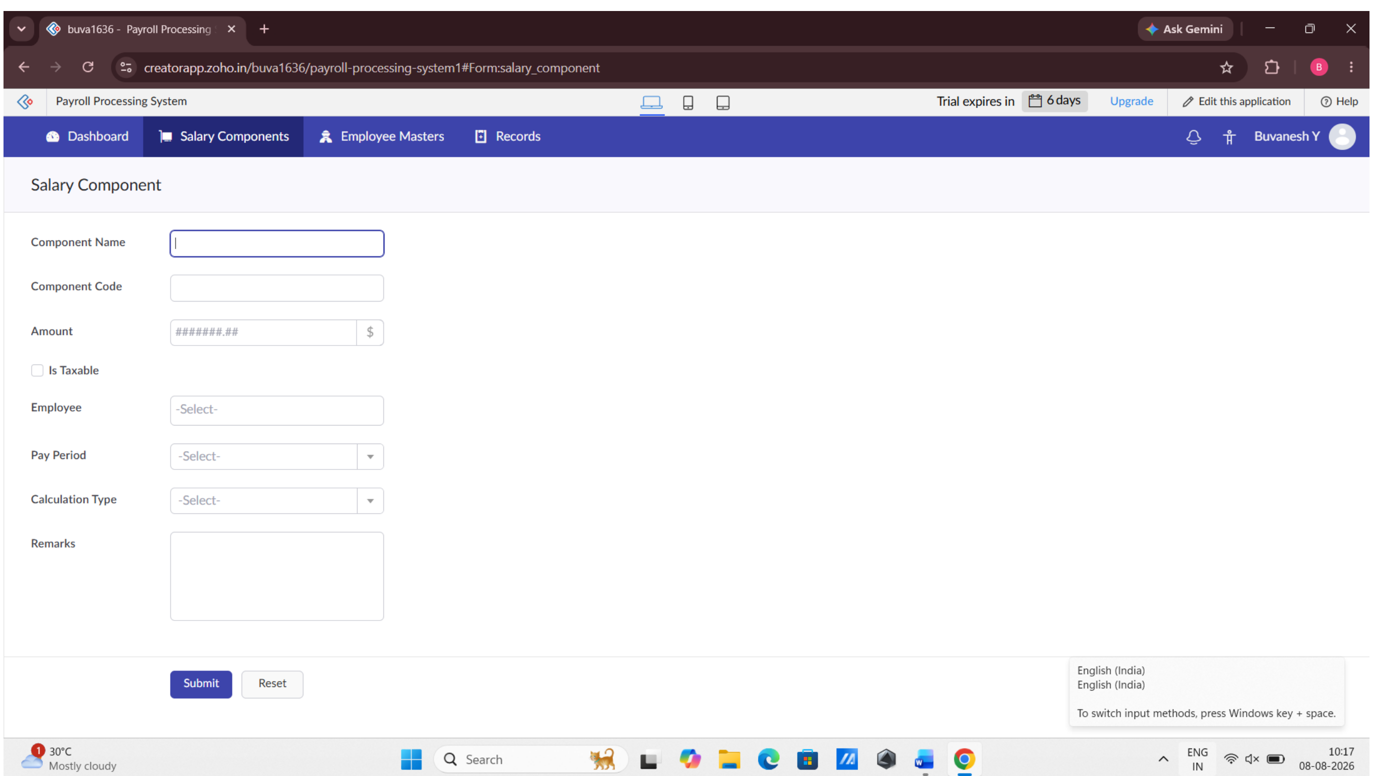
STEP 5: CREATED NEW APPLICATION

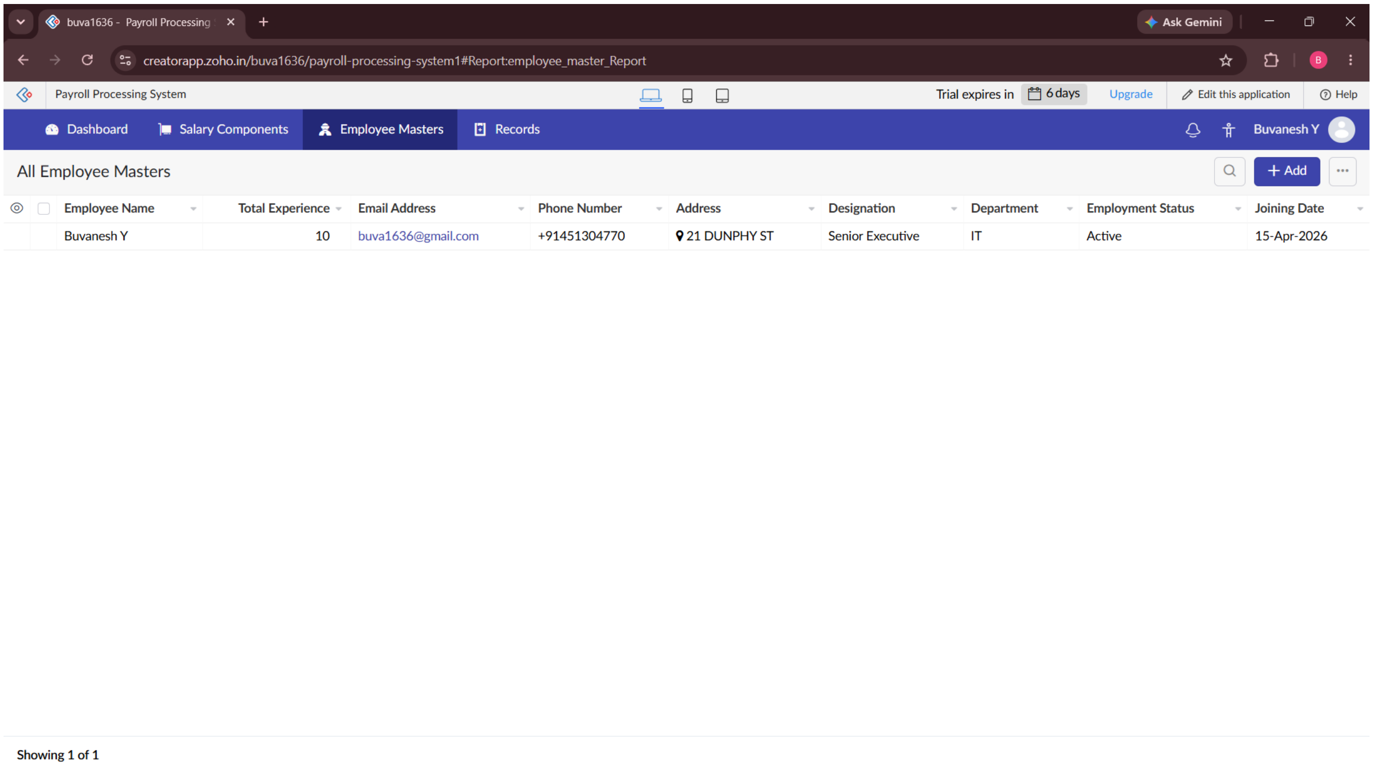


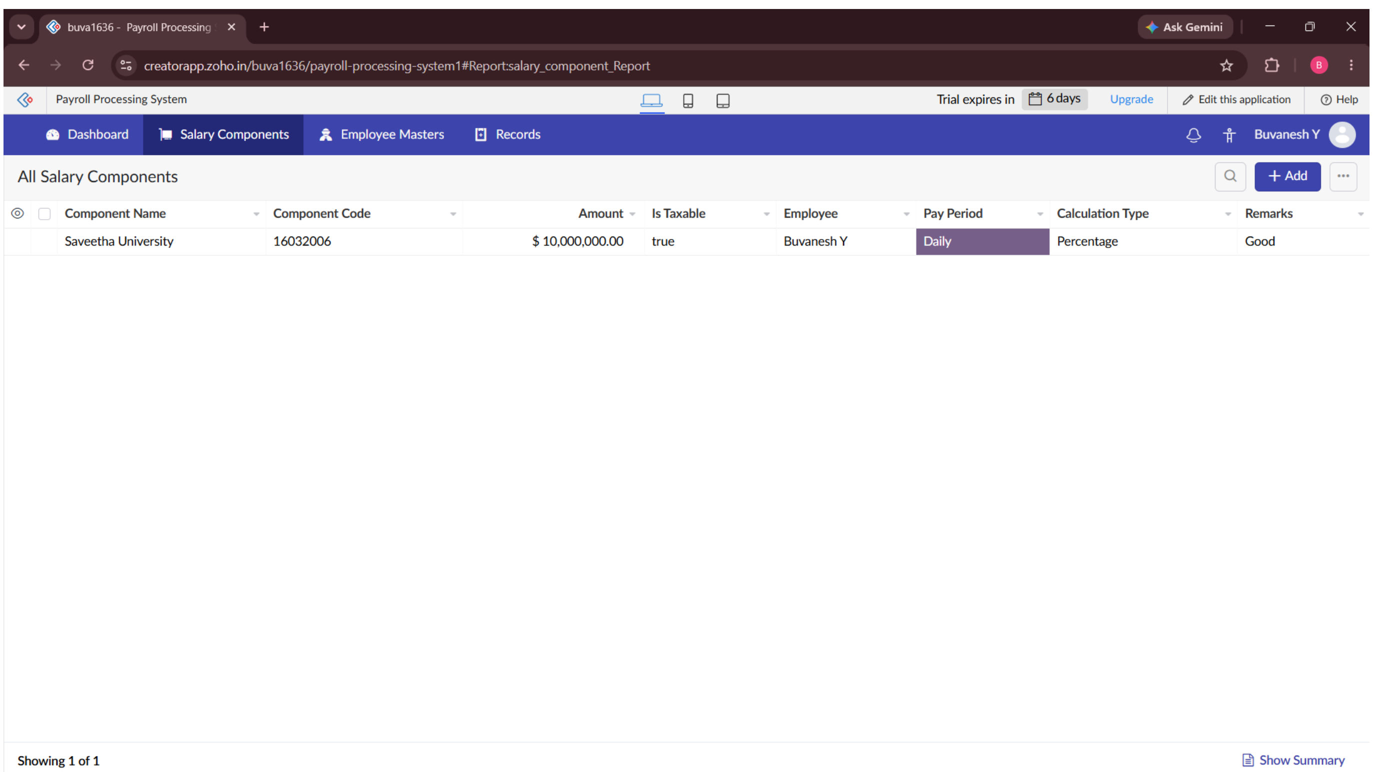
❖ Create Multiple Forms according to our application.

STEP 6: THE SOFTWARE HASE BEEN CREATED.









RESULT:

Thus, the **Payroll Processing System** was successfully created using **Zoho Creator** as a cloud service provider, demonstrating the **SaaS model**. The application allows employee salary details to be stored, processed, and accessed through the cloud.

EXP NO 3 :Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

DATE:

AIM:

To demonstrate virtualization by installing type-2 hypervisor in your device, create and configure VM image with a host operating system (either windows/linux).

PROCEDURE:

STEP 1:Dowload VMware workstation and installed as type 2hypervisor.

STEP2:Dowload ubuntu or tiny OS as iso image file.

STEP 3: In VMware workstation->create new VM.

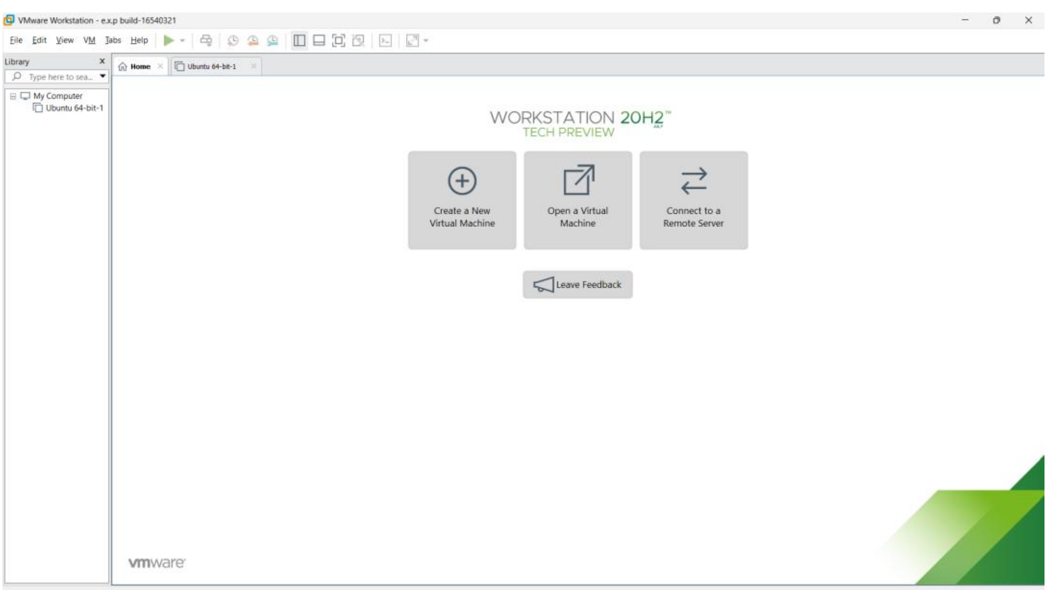
STEP 4: Do the basic configuration settings.

STEP 5: Created tiny OS virtual machine.

STEP 6: Launch the VM.

IMPLEMENTATION:

STEP 1:DOWLOAD VMWARE WORKSTATIONAND INSTALLED AS TYPE 2HYPERVISOR

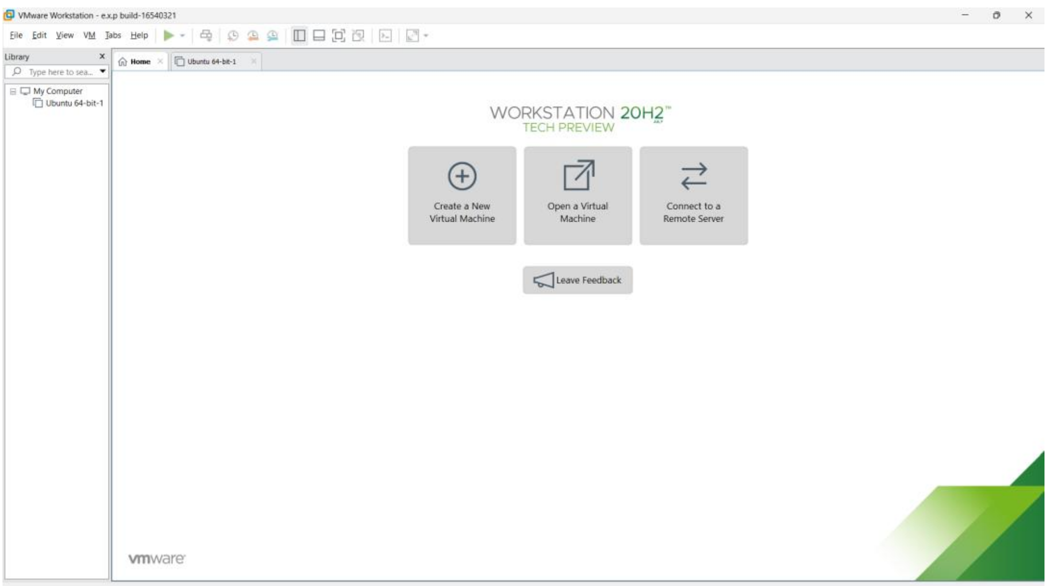


STEP2: DOWLOAD UBUNTU OR TINY OS AS ISO IMAGE FILE

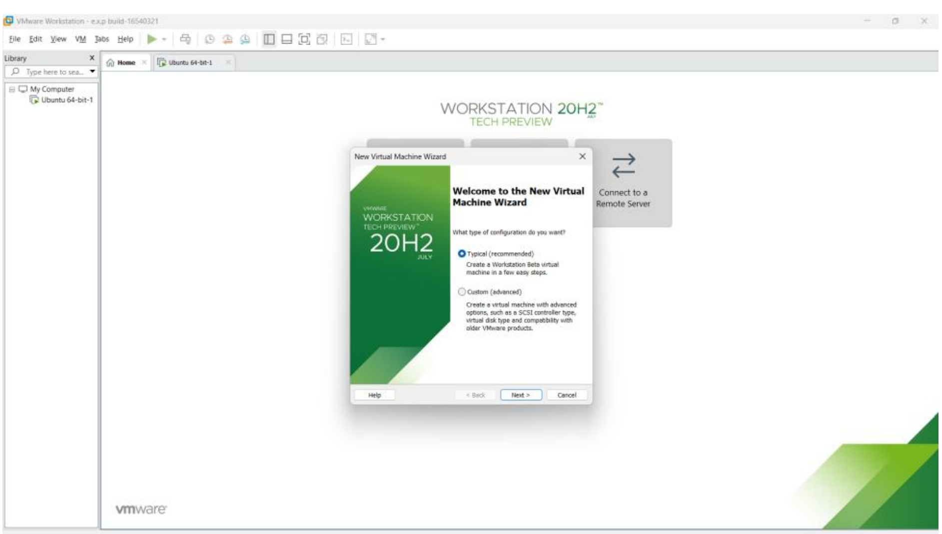
Index of /11.x/x86/release/

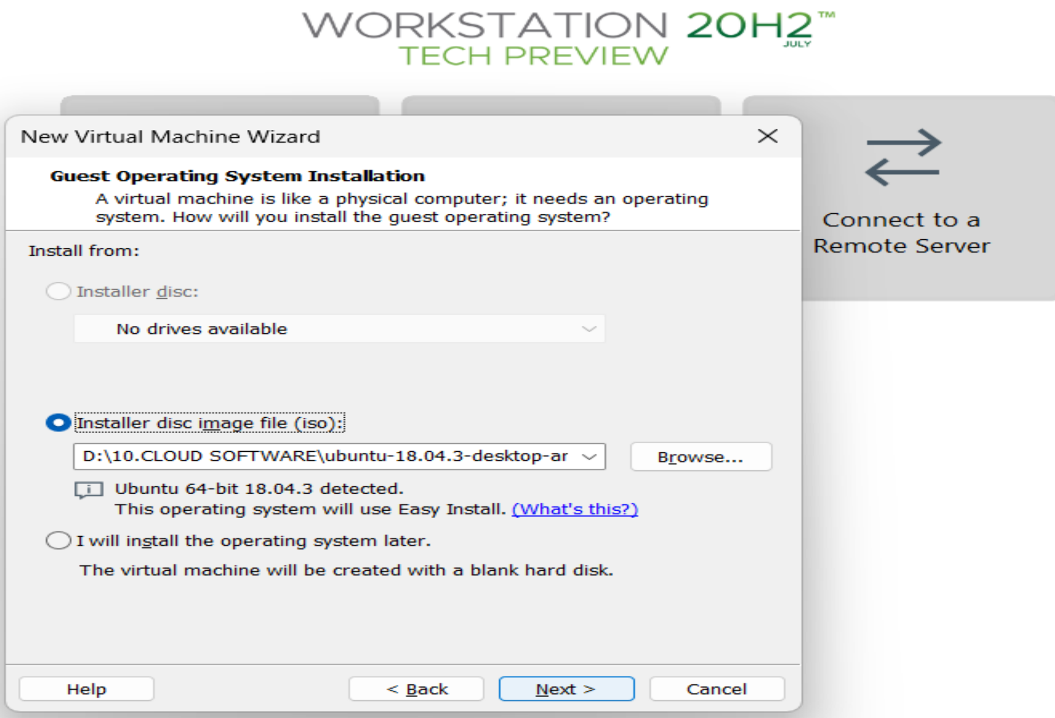
../		
distribution_files/	09-Feb-2020 11:50	-
src/	03-Dec-2019 11:14	-
Core-11.1.iso	01-Apr-2020 07:49	14757888
Core-11.1.iso.md5.txt	01-Apr-2020 07:49	48
Core-11.1.iso.zsync	01-Apr-2020 07:49	50639
Core-current.iso	01-Apr-2020 07:49	14757888
CorePlus-11.1.iso	01-Apr-2020 07:50	216006656
CorePlus-11.1.iso.md5.txt	01-Apr-2020 07:50	52
CorePlus-11.1.iso.zsync	01-Apr-2020 07:50	369358
CorePlus-current.iso	01-Apr-2020 07:50	216006656
TinyCore-11.1.iso	01-Apr-2020 07:50	19922944
TinyCore-11.1.iso.md5.txt	01-Apr-2020 07:50	52
TinyCore-11.1.iso.zsync	01-Apr-2020 07:50	68301
TinyCore-current.iso	01-Apr-2020 07:50	19922944

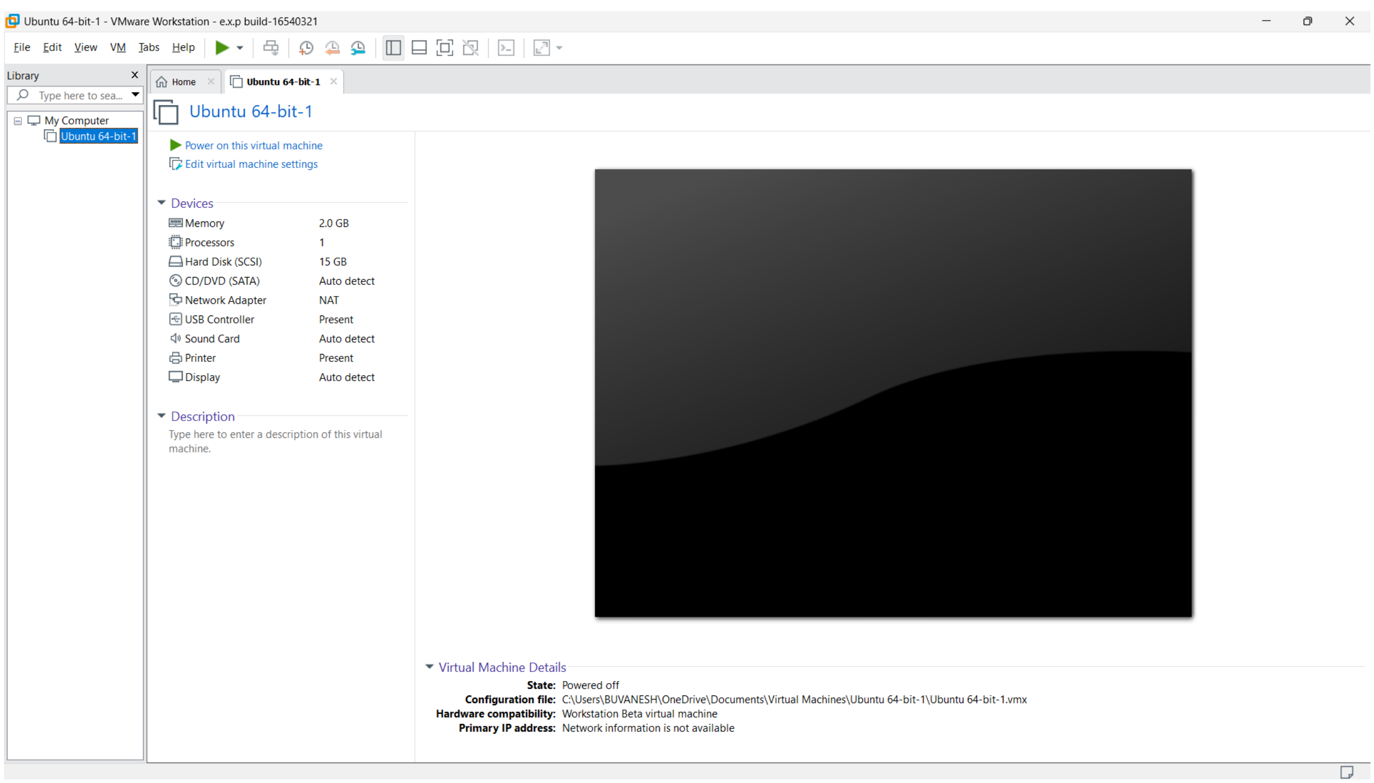
STEP 3: IN VMWARE WORKSTATION->CREATE NEW VM

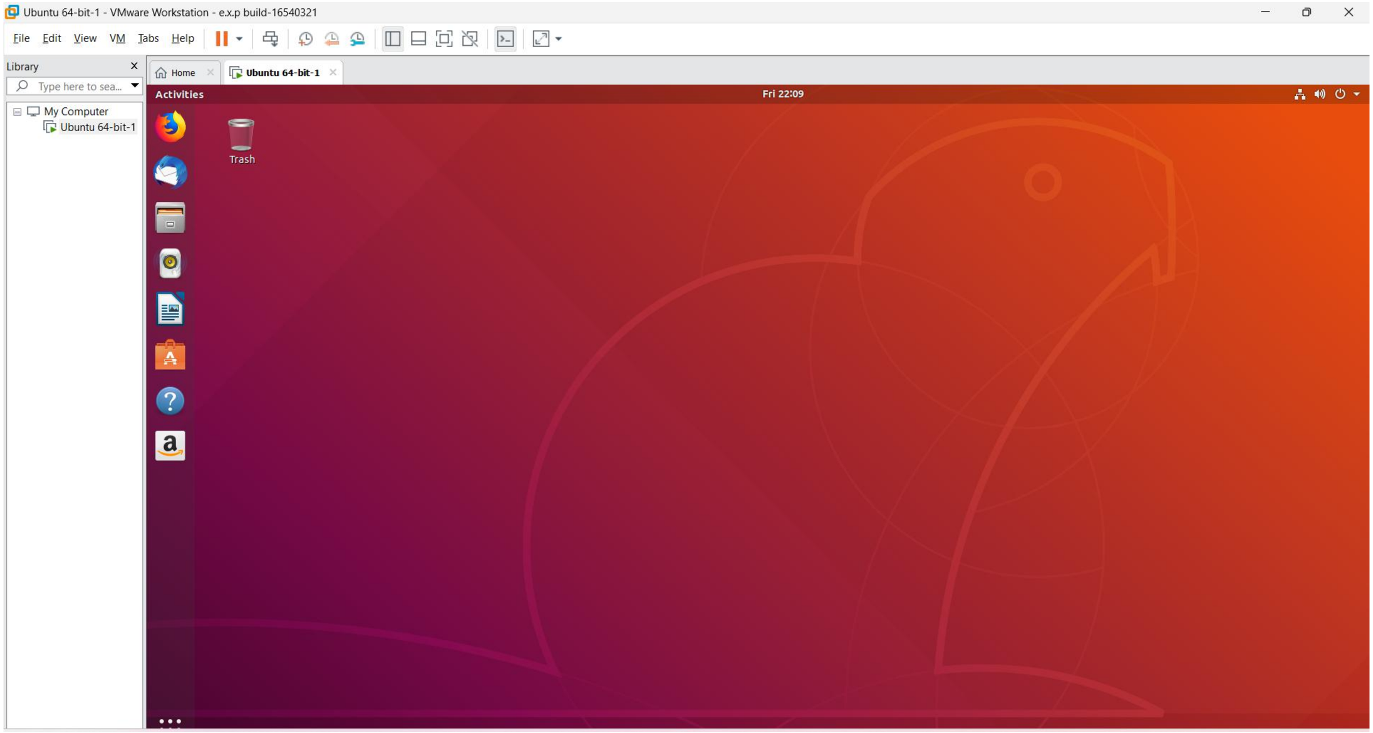


STEP 4: DO THE BASIC CONFIGURATION SETTINGS.




WORKSTATION 20H2™
JULY
TECH PREVIEW
New Virtual Machine Wizard
Guest Operating System Installation
A virtual machine is like a physical computer; it needs an operating system. How will you install the guest operating system?
Install from:
Installer disc:
No drives available
Installer disc image file (iso):
D:\10.CLOUD SOFTWARE\ubuntu-18.04.3-desktop-ar
Browse...
Ubuntu 64-bit 18.04.3 detected.
This operating system will use Easy Install. (What's this?)
I will install the operating system later.
The virtual machine will be created with a blank hard disk.
Help
< Back
Next >
Cancel
Connect to a Remote Server


Ubuntu 64-bit-1 - VMware Workstation - e.x.p build-16540321
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
Ubuntu 64-bit-1
Home
Ubuntu 64-bit-1
Ubuntu 64-bit-1
Power on this virtual machine
Edit virtual machine settings
Devices
Memory 2.0 GB
Processors 1
Hard Disk (SCSI) 15 GB
CD/DVD (SATA) Auto detect
Network Adapter NAT
USB Controller Present
Sound Card Auto detect
Printer Present
Display Auto detect
Description
Type here to enter a description of this virtual machine.
Virtual Machine Details
State: Powered off
Configuration file: C:\Users\BUVANESH\OneDrive\Documents\Virtual Machines\Ubuntu 64-bit-1\Ubuntu 64-bit-1.vmx
Hardware compatibility: Workstation Beta virtual machine
Primary IP address: Network information is not available


Ubuntu 64-bit-1 - VMware Workstation - e.x.p build-16540321
File
Edit
View
VM
Tabs
Help
Library
Type here to sea...
My Computer
Ubuntu 64-bit-1
Home
Ubuntu 64-bit-1
Activities
Fri 22:09
Trash

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

AIM:

To create a snapshot of a vm and test it by loading the previous version/cloned vm

PROCEDURE:

STEP 1: GOTO VMWARE WORKSTATION.

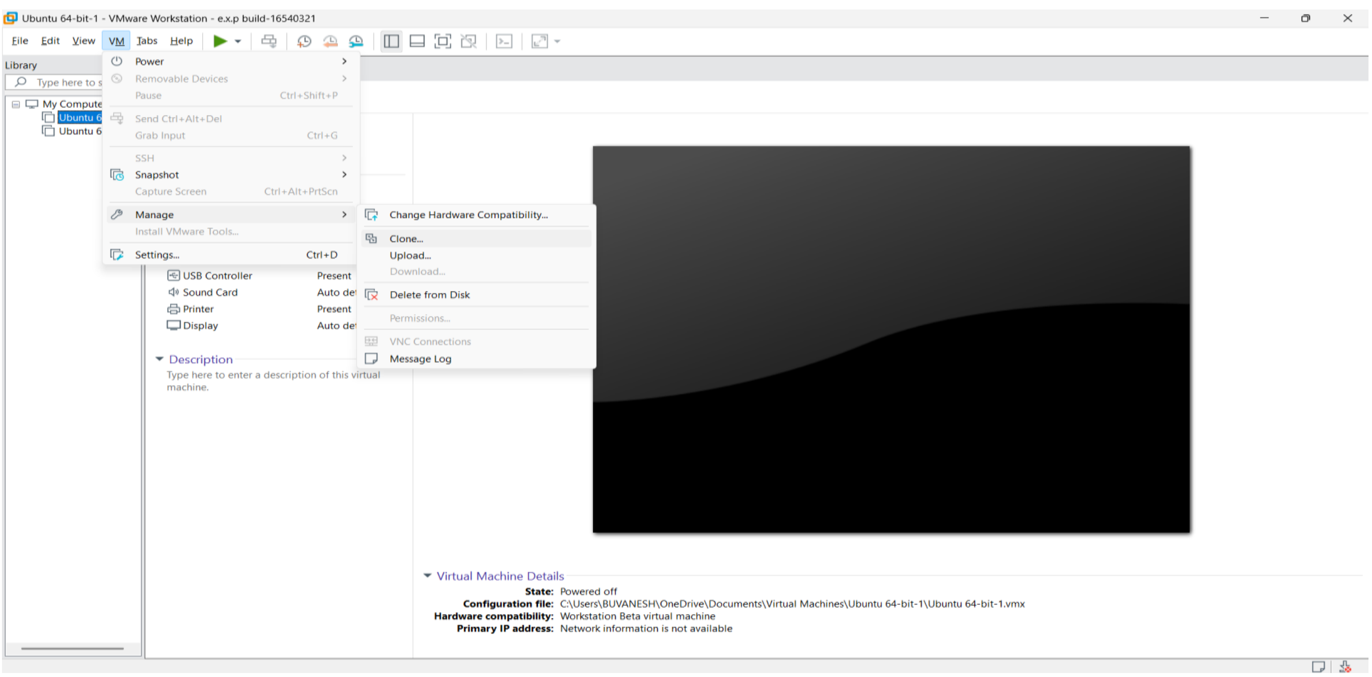
STEP 2: CREATE FILES ON DESKTOP.

STEP 3: CLICK ON VM AND SELECTS SNAPSHOT-> TAKE SNAPSHOT.

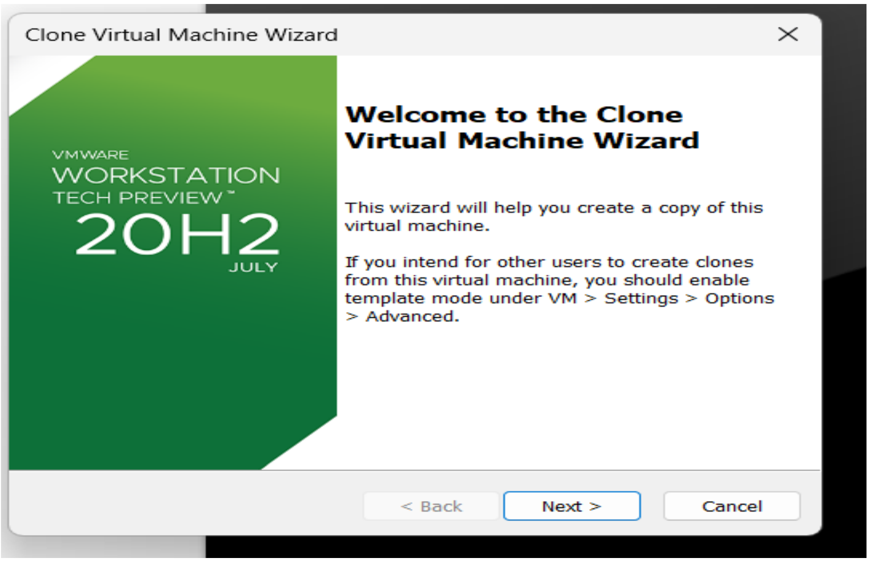
STEP 4: SNAPSHOT IS BEING DONE

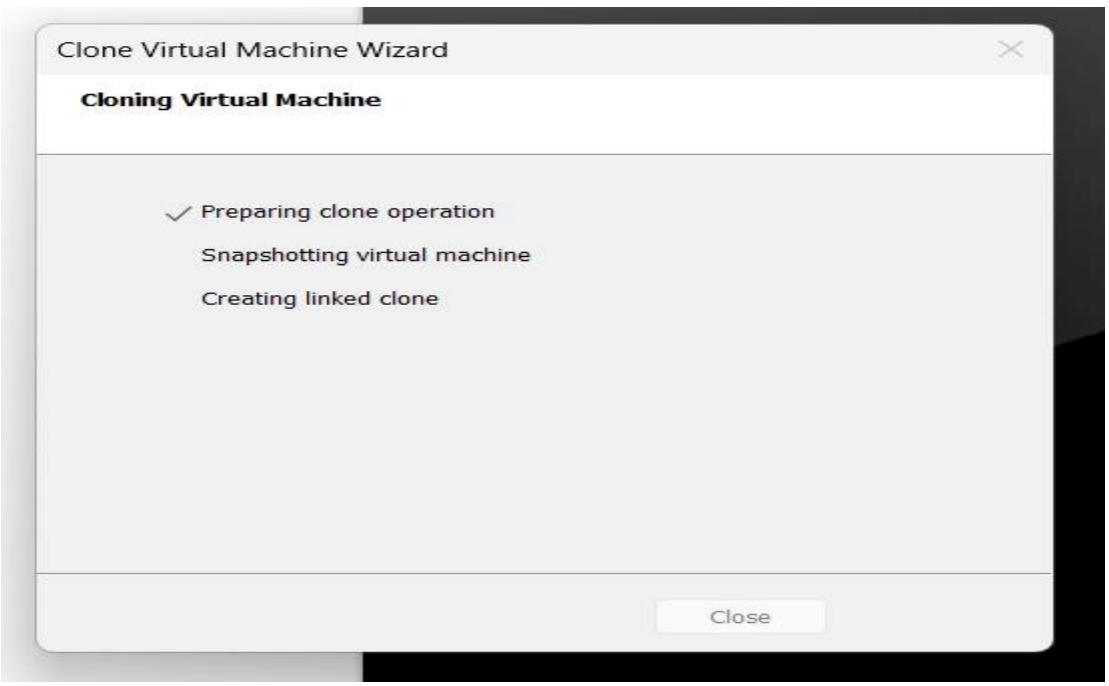
IMPLEMENTATION:

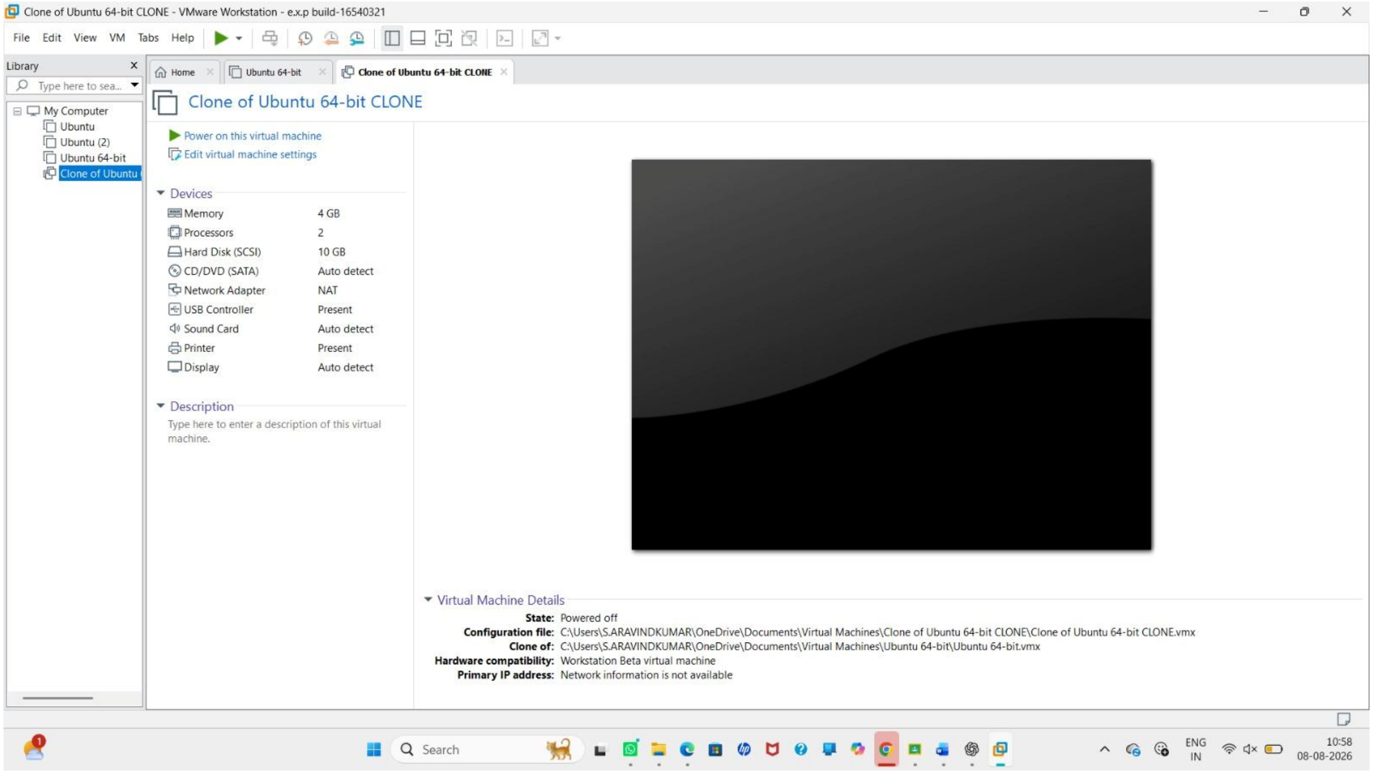
STEP 1: GO TO VM AND GOTO MANAGE AND CLICK CLONE



STEP 2: CLICK CLONE






Clone of Ubuntu 64-bit CLONE - VMware Workstation - e.x.p build-16540321
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
Ubuntu
Ubuntu (2)
Ubuntu 64-bit
Clone of Ubuntu
Home
Ubuntu 64-bit
Clone of Ubuntu 64-bit CLONE
Clone of Ubuntu 64-bit CLONE
Power on this virtual machine
Edit virtual machine settings
Devices
Memory 4 GB
Processors 2
Hard Disk (SCSI) 10 GB
CD/DVD (SATA) Auto detect
Network Adapter NAT
USB Controller Present
Sound Card Auto detect
Printer Present
Display Auto detect
Description
Type here to enter a description of this virtual machine.
Virtual Machine Details
State: Powered off
Configuration file: C:\Users\S.ARAVINDKUMAR\OneDrive\Documents\Virtual Machines\Clone of Ubuntu 64-bit CLONE\Clone of Ubuntu 64-bit CLONE.vmx
Clone of: C:\Users\S.ARAVINDKUMAR\OneDrive\Documents\Virtual Machines\Ubuntu 64-bit\Ubuntu 64-bit.vmx
Hardware compatibility: Workstation Beta virtual machine
Primary IP address: Network information is not available
Search
ENG
IN
10:58
08-08-2026